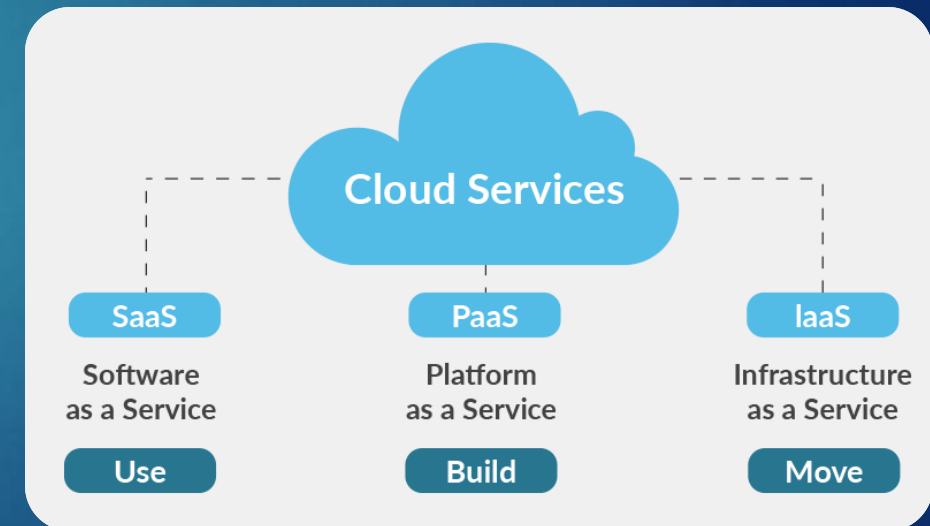




- ▶ Sign up for an AWS “Paid Plan” account
 - ▶ You will need this today!
- ▶ Team registration due by midnight 9/10
 - ▶ Remember to use Slack channel [#project-team-formation](#) to look for a team or if you are a team in need of people

- ▶ Service Models (SaaS, PaaS or IaaS)?
 - ▶ Fred is working with the cloud and notices that the cloud provider takes responsibility for the operating system, and he needs to assume responsibility for any apps or services running on the operating system. Which service model is he operating under?
 - ▶ Which service model requires you to just only the users and data?
 - ▶ Which service model provides the greatest amount of flexibility?

- ▶ Software as a Service (SaaS)
 - ▶ Offers purpose-built business applications on a subscription basis
- ▶ Platform as a Service (PaaS)
 - ▶ Provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure
- ▶ Infrastructure as a Service (IaaS)
 - ▶ Offers storage and compute resources that developers and IT organizations use to deliver custom business solutions
 - ▶ Customers rent computing resources vs. buying and installing them in their own data center



Organizations moving to the cloud – SaaS, PaaS or IaaS ?

Some questions to ask (from last class):

- ▶ What services do you want running in the cloud?
- ▶ What are your security requirements?
- ▶ What is your budget?
- ▶ What is the total # of users?
- ▶ Lift and shift or rebuild?
- ▶ Existing partnerships?
- ▶ What does your dev team look like (how much experience)
- ▶ Time to market?
- ▶ How do you want to support your applications?
- ▶ What's your motivation to move to the cloud?
- ▶ What does current architecture look like?
- ▶ Any regulations that need to be considered?

Virtualization in Cloud Computing

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology



Economics of Cloud Computing

- ▶ Cloud computing has several economic motivations for organizations
 1. Utility-based pricing
 2. Reduction in capital expenditures
 3. Improved resource utilization and allocation
- ▶ The common thread across these benefits is the ability to share computing infrastructure at scale, which was not feasible with traditional on-premises systems
- ▶ None of this would be possible without virtualization technology



#1 - Utility-based pricing

- ▶ Cloud services don't need to be the lowest-cost option to be cost-effective. The value depends on **how and when** the service is used.
- ▶ Example: Transportation options:
 - ▶ Buy a car: \$500/month (\$17/day)
 - ▶ Rent a car: \$40/day
 - ▶ Use Uber: \$20/ride
- ▶ If you only need a car 2 days per week, buying becomes more expensive:
 - ▶ Renting: 2 days/week * \$40 = \$80/week → \$320/month (save \$180/month compared to owning)
 - ▶ Renting also removed overhead costs like maintenance, repairs and insurance
- ▶ Whether a solution is economical depends on **usage patterns**, not just price

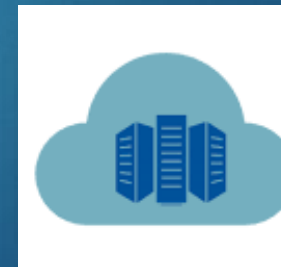
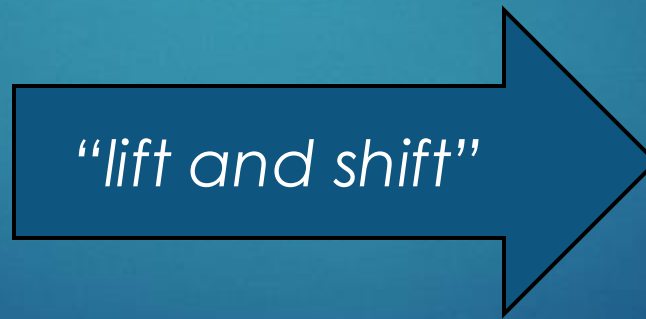


Datacenters (aka on-premise) vs. Cloud

- ▶ Consider this:
 - ▶ Operating a large data center can cost \$10-\$25 million per year to run
 - ▶ An organization decides to transfer all their infrastructure from their data center to the cloud in an effort to save costs
 - ▶ This is sometimes referred to as a “lift and shift”, since all hardware is moved (i.e. lifted) and replicated as-is (i.e. shifted) in the cloud



Traditional
Data Center



Cloud

Question: Does the organization save \$\$\$?

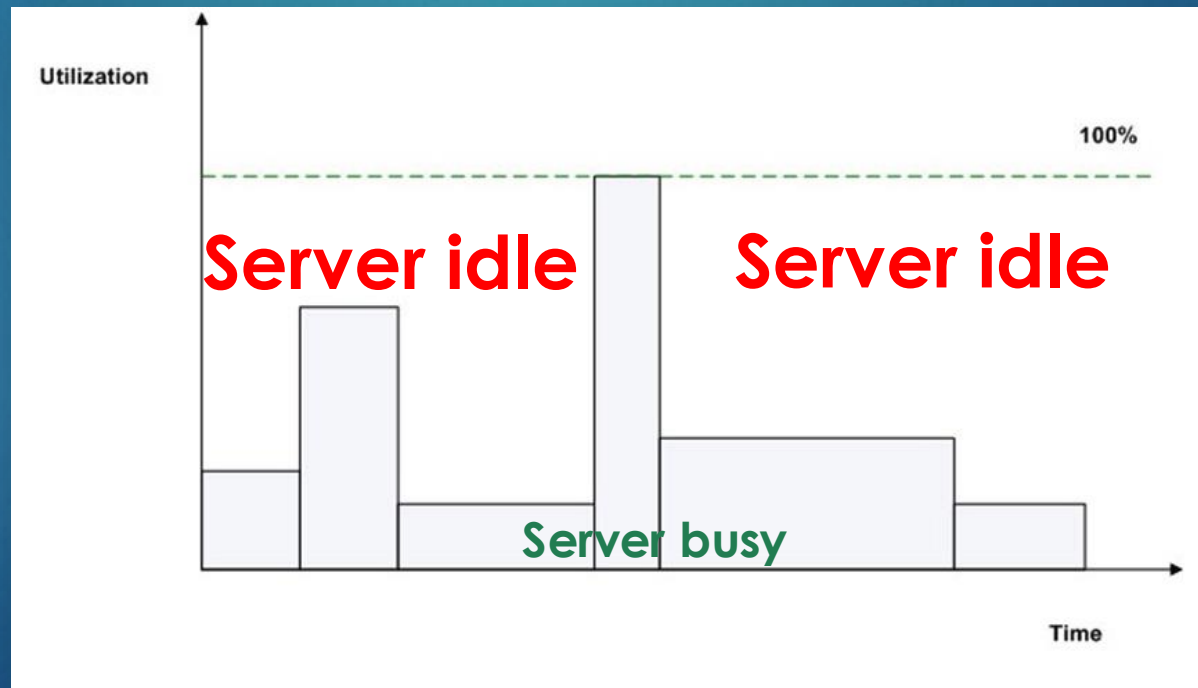
Datacenters vs. Cloud

- ▶ Answer: Probably not...Why?
- ▶ It estimated that more than 80% of on-premise enterprise workloads are overprovisioned, meaning they have more compute power, memory and storage than they will ever need to operate
- ▶ By retaining that excess capacity as these workloads are moved to the cloud, companies can actually increase the costs of running them by up to 15%
- ▶ Instead of becoming more efficient, they're merely transferring their existing inefficiencies to a new location (and paying more)



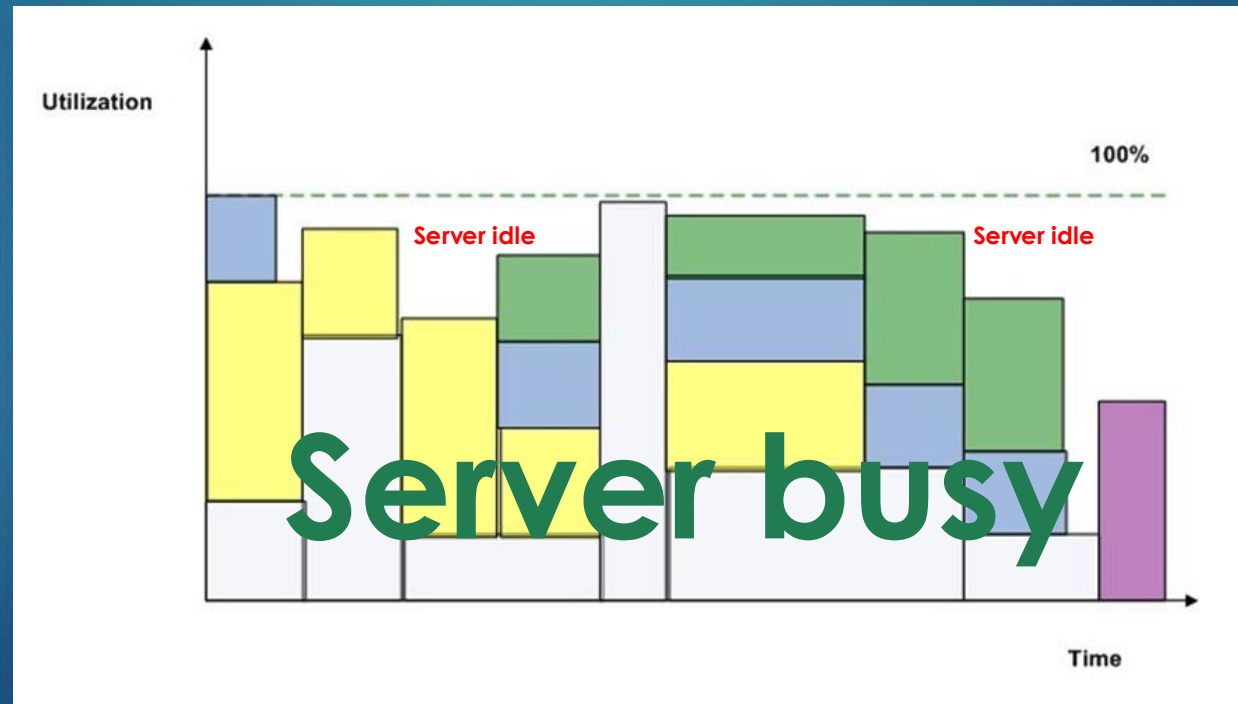
Dedicated Server Utilization

- ▶ Physical servers with fixed configurations are less likely to have high-utilization rates
- ▶ Unless there is a steady stream of jobs that fits the machine's configuration, there will be idle periods



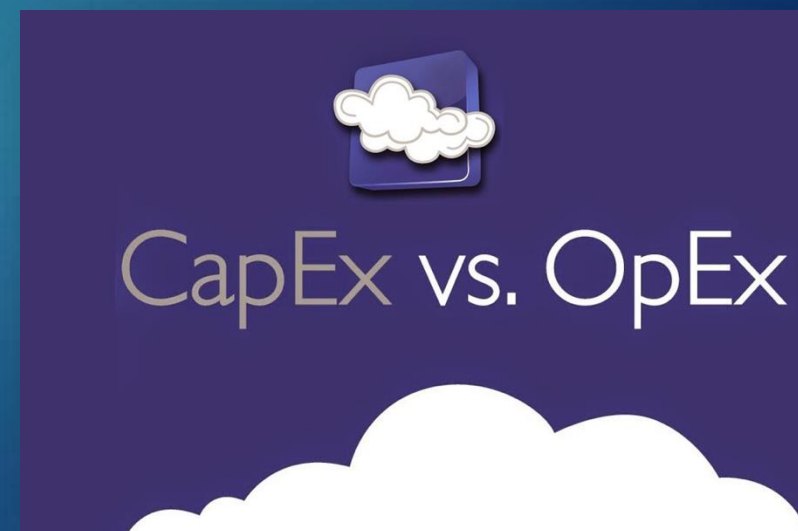
Shared Server Utilization

- ▶ By repurposing the physical servers with virtual machine images, a single server can now function as multiple servers
- ▶ This allows the ability to run more applications simultaneously and leveraging the server's capacity much more efficiently



#2 – Reduction in Capital Expenditures

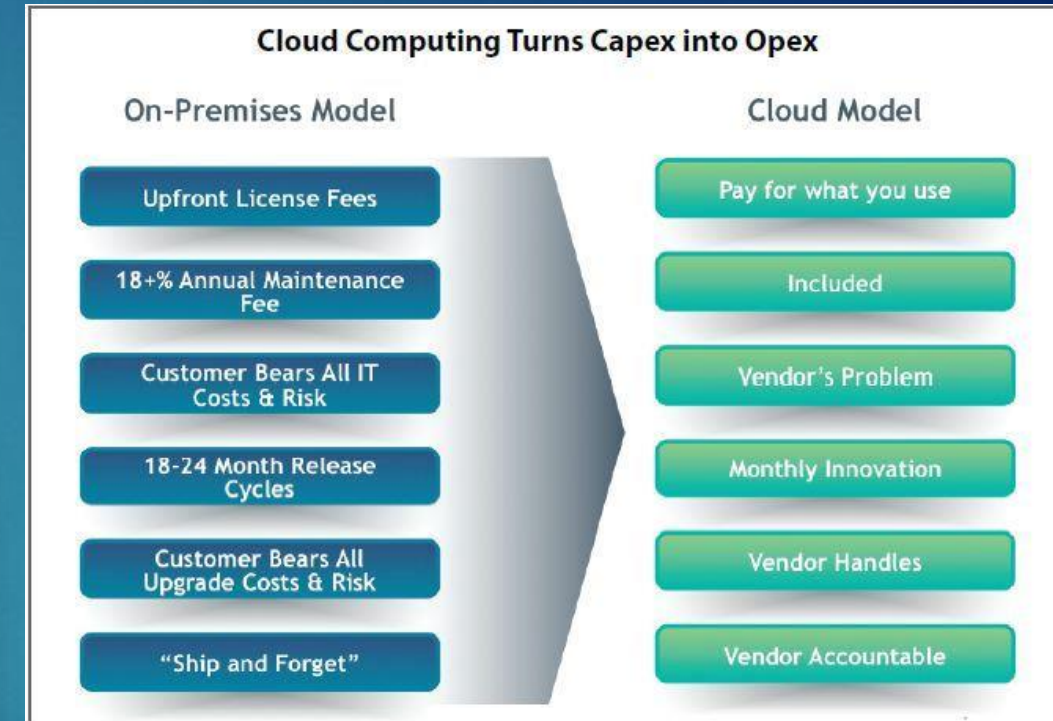
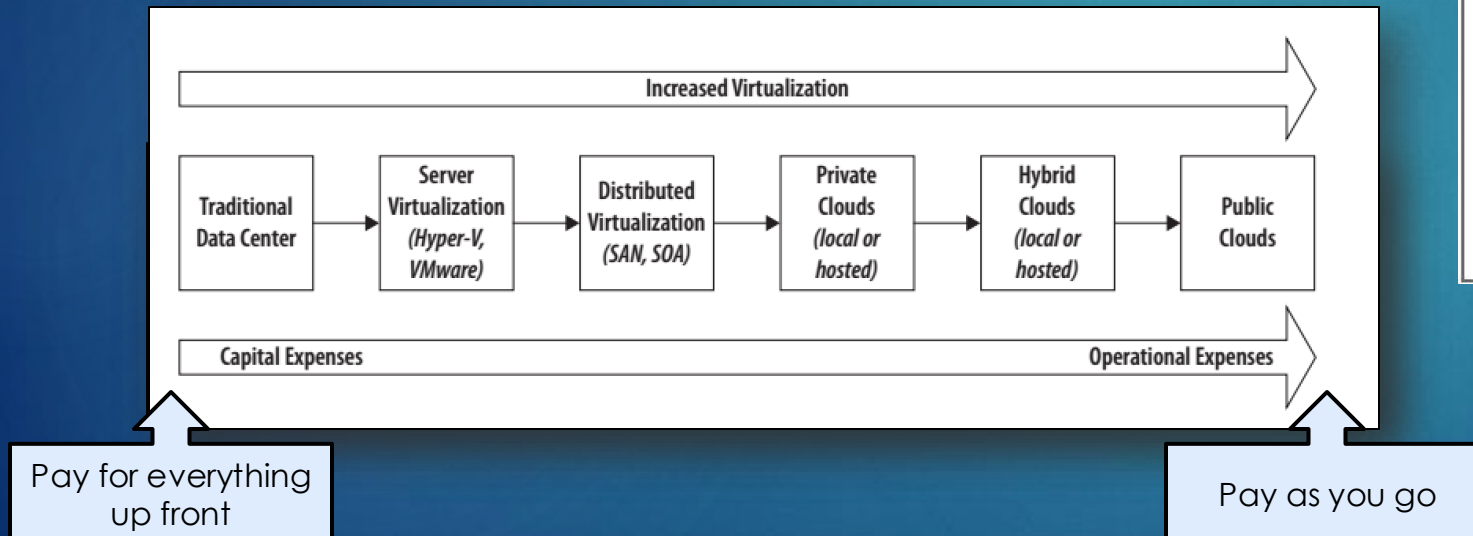
- ▶ Cloud computing changes the ratio between capital expenses (CAPEX) and operational expenses (OPEX)
- ▶ CAPEX:
 - ▶ Requires a large up-front investment to be amortized (written down) as its value decreases over time
 - ▶ The initial CAPEX purchase can require a large cash outlay, which is a barrier for many startups
- ▶ OPEX:
 - ▶ It's a day-to-day expense which can much more easily be increased or decreased as business needs dictate



#2 – Reduction in Capital Expenditures

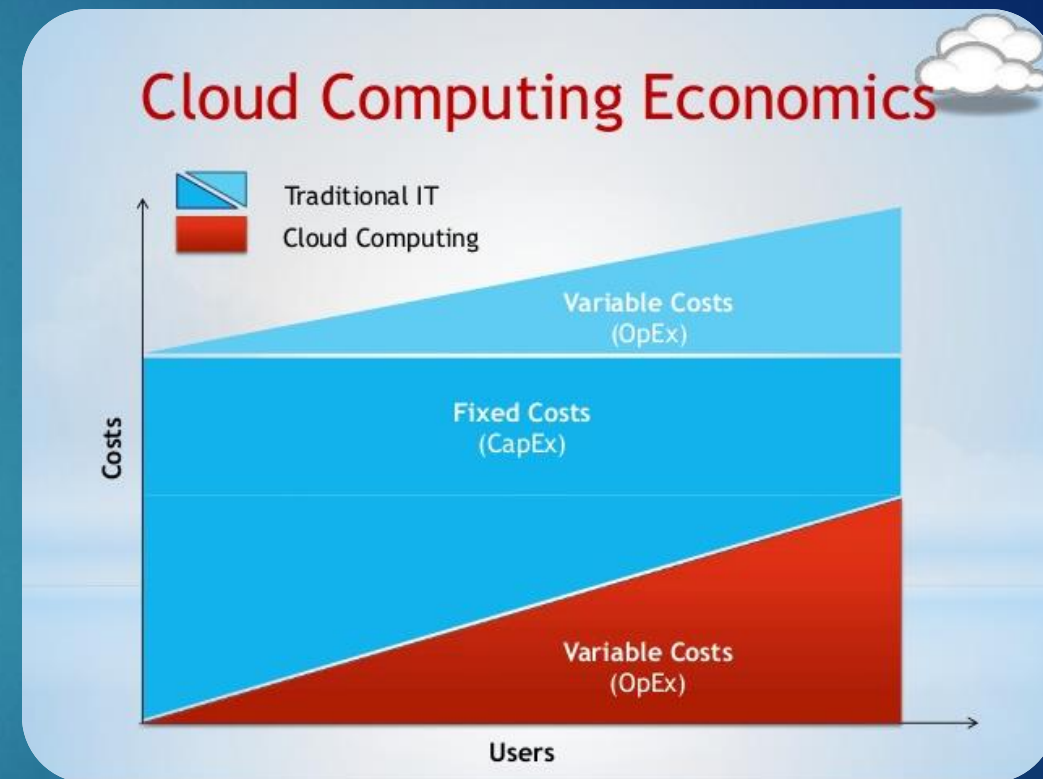
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- ▶ As organizations migrate from their data center to the cloud, CAPEX is reduced, and virtualization is increased



#2 – Reduction in Capital Expenditures

- ▶ Pay-as-you-go pricing shifts costs from upfront capital to ongoing operational expenses
- ▶ Lower initial barriers make it easier to start projects without heavy infrastructure investment
- ▶ Costs scale with usage, staying low until demand for the application increases significantly



#3 - Improved resource utilization and allocation

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Traditional
Data Center

VS.

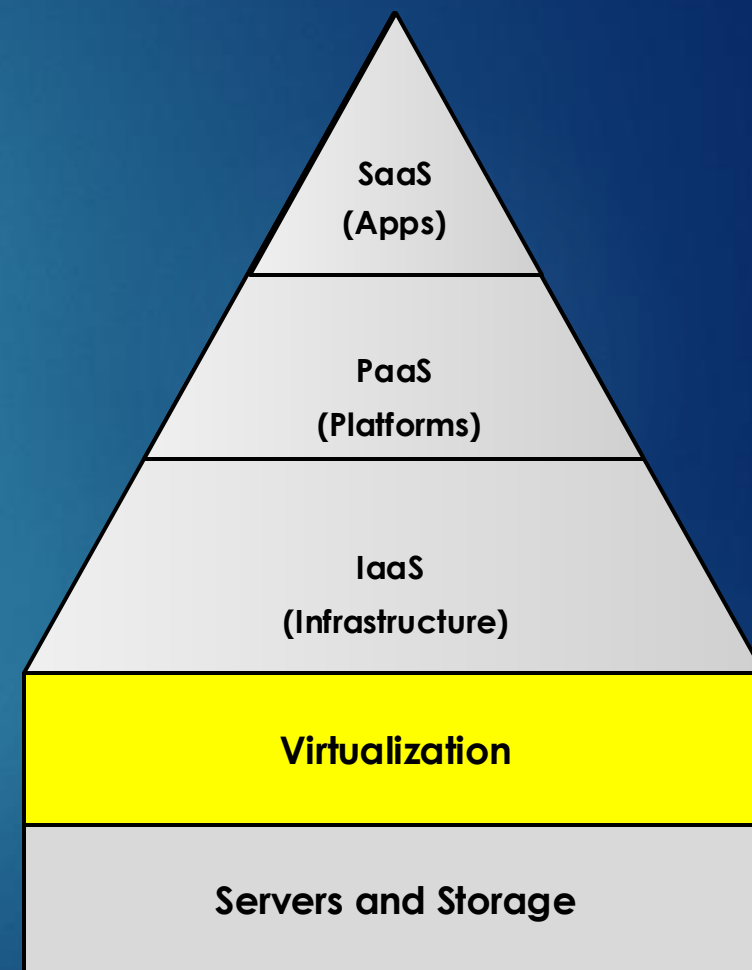


Cloud

- ▶ Must purchase hardware upfront for anticipated “peak” performance
 - ▶ Must plan for disaster recovery separately at the department or project level
 - ▶ Stand-by server would be needed to act as a backup for each primary server
 - ▶ Each system must be individually patched to keep up to date with security and performance patches
- ▶ Ability to manage workloads and allocate jobs to available servers through the use of rapidly deployed virtual machine images to servers with excess capacity
 - ▶ Clouds can be configured to use geographically distributed data centers and replication services between the data centers to provide disaster recovery for all cloud consumers
 - ▶ If a server were to fail in the cloud, it could simply be removed from the pool of available resources; jobs would continue to run on other servers
 - ▶ Virtual machine images can be patched and when new instances of virtual machines are started, the patched images are deployed

Virtualization

- ▶ Involves creating a virtual version of physical resources such as servers, storage, and networks
- ▶ Separates compute and storage services from the underlying physical hardware
- ▶ Enables better resource utilization, scalability, and flexibility across workloads
- ▶ Serves as the core foundation that enables cloud computing



Virtualization - Characteristics

▶ Partitioning

- ▶ Run multiple operating systems (Windows, Linux, Mac) on one physical machine

▶ Isolation

- ▶ Each virtual machine is isolated from its host physical system and other virtualized machine
- ▶ If one virtual instance crashes, it doesn't affect the other virtual machines

▶ Encapsulation

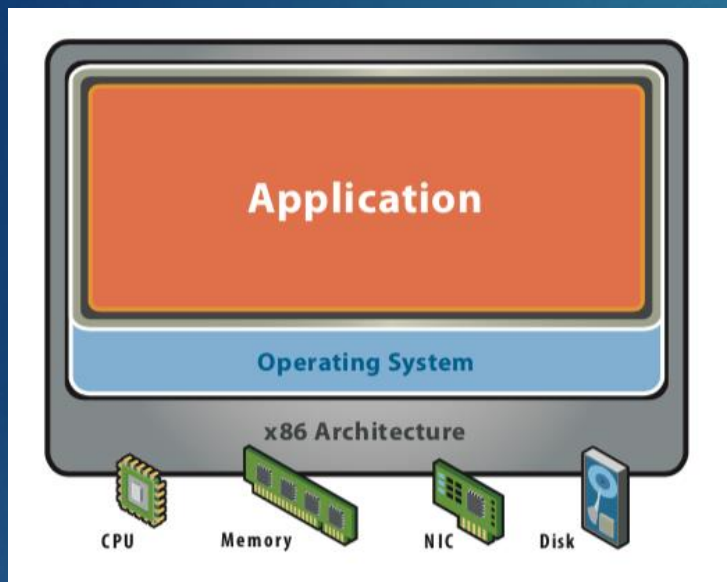
- ▶ Save the entire state of a virtual machine to files
- ▶ Move and copy virtual machines as easily as moving and copying files

▶ Hardware Independence

- ▶ Provision or migrate any virtual machine to any physical server

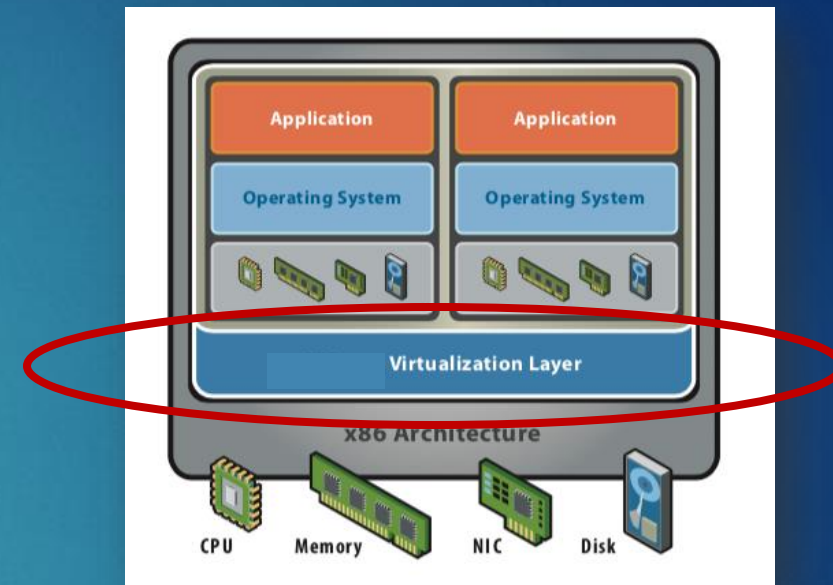
Virtualization – Before & After

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Before Virtualization:

- ▶ Single OS per machine
- ▶ Software and hardware tightly coupled
- ▶ Running multiple applications on same machine often creates conflict
- ▶ Underutilized resources
- ▶ Inflexible and costly infrastructure



After Virtualization:

- ▶ Hardware-independence of operating system and applications
- ▶ Virtual machines can be provisioned to any system
- ▶ Can manage OS and application as a single unit by encapsulating them into virtual machines

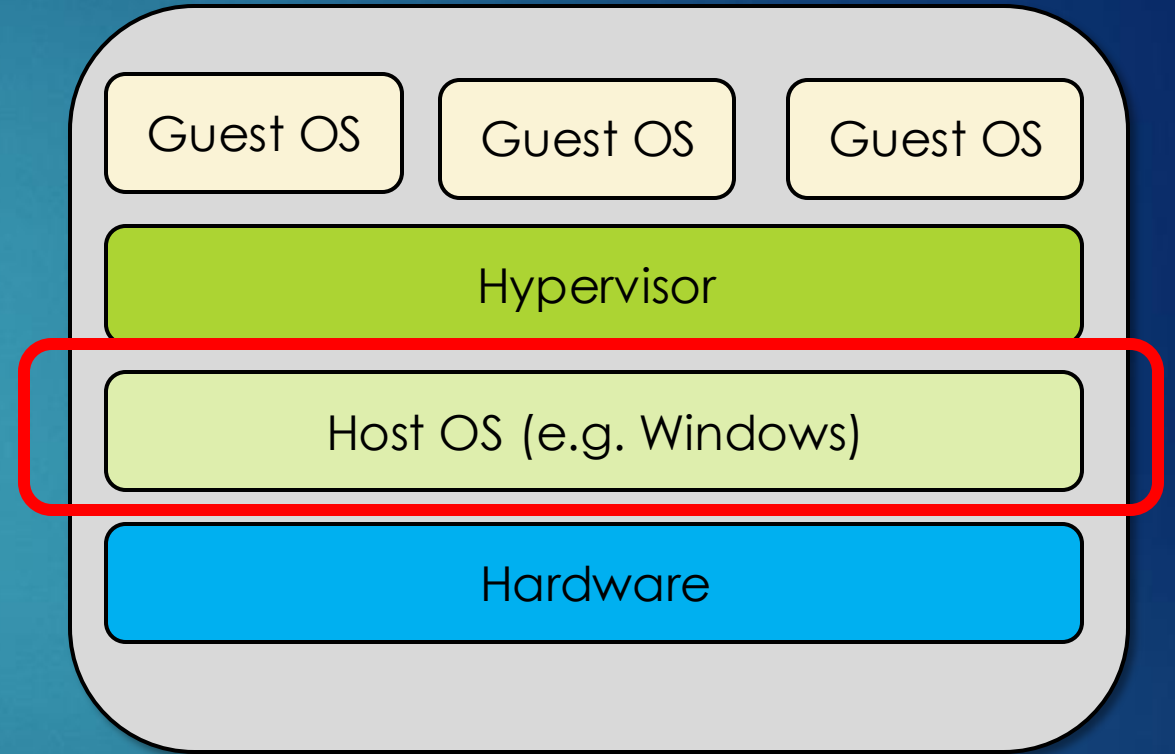
Virtualization Layer = Hypervisor

- ▶ A Hypervisor, a.k.a. a virtual machine manager/monitor (VMM), is a program that allows multiple operating systems to share a single hardware host
- ▶ It knows how to act as a “traffic cop” to make things happen in an orderly manner
 - ▶ It sits at the lowest levels of the hardware environment
- ▶ Two major types: Hosted and Native



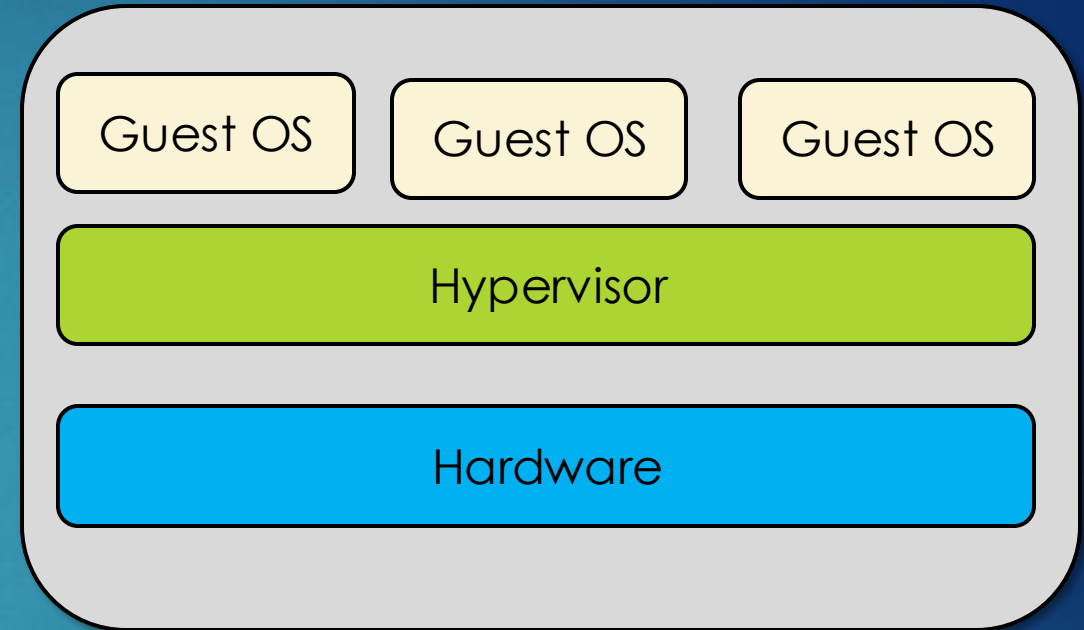
Hypervisor - Hosted

- ▶ Require a host OS
- ▶ Can run multiple OS but doesn't have access to hardware
- ▶ More overhead
- ▶ If host crashes, you lose access to guest OS
- ▶ Examples: VMWare Workstation, VMware Parallels, Oracle Virtualbox, and Microsoft VirtualPC



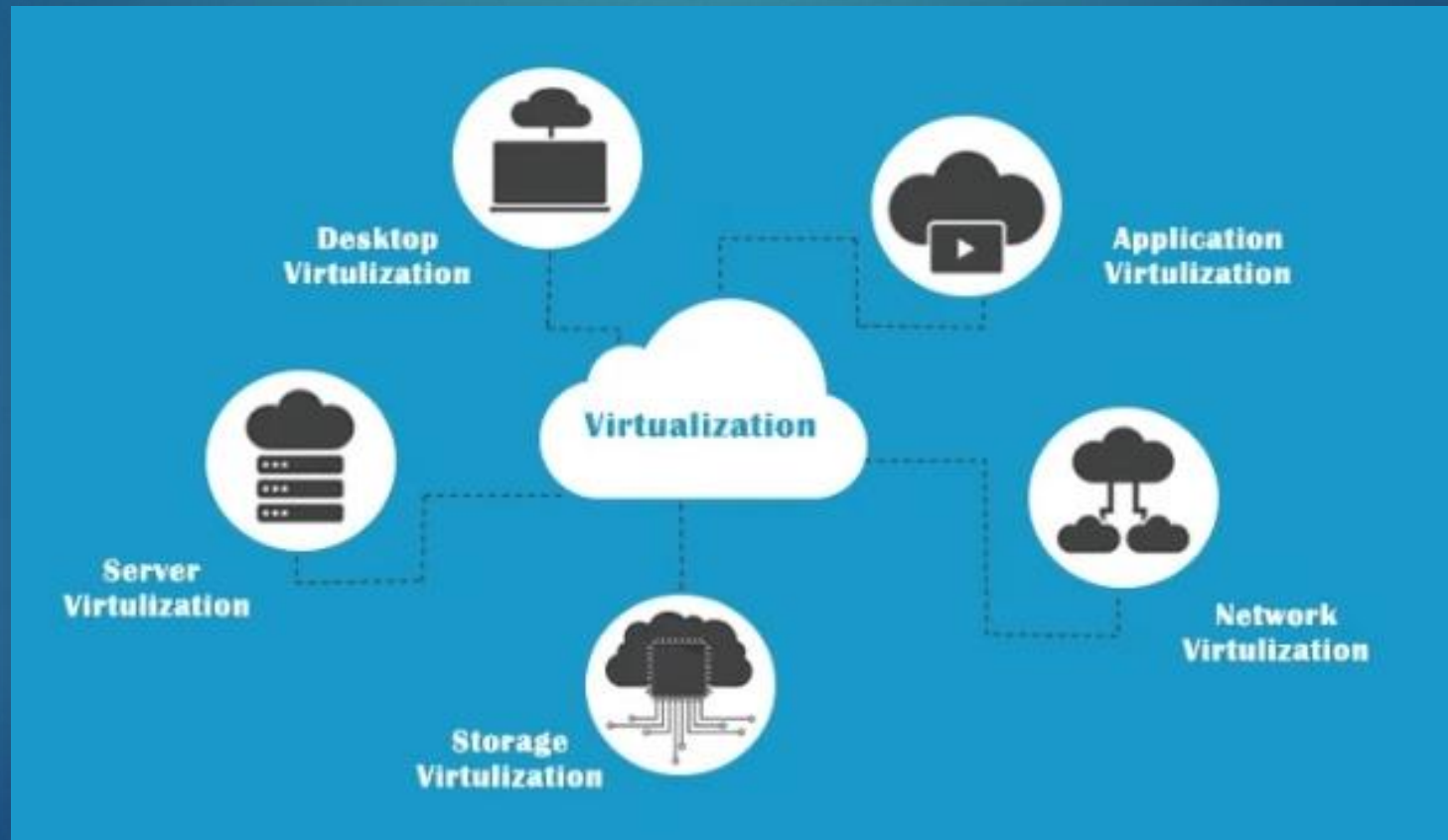
Hypervisor – Native

- ▶ No hosting OS
- ▶ Installs directly onto a computer
- ▶ Hypervisor has direct access to all hardware and features
- ▶ Used for servers because of their security and portability to move from hardware to hardware in case of a crash
- ▶ Examples: VMware ESXi, Citrix XenServer, and Microsoft Hyper-V



What else can Virtualized?

- ▶ Virtualization can be applied very broadly to just about everything that you could imagine:



Virtualization - Advantages

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Server consolidation

- Reduces hardware, power, and maintenance costs by running multiple virtual servers on a single physical host
- Simplifies server management and minimizes physical infrastructure footprint

Infrastructure Optimization

- Dynamically allocates compute, memory, and storage based on workload demands
- Enables resource scaling without disrupting other running virtual machines

Flexibility and Scalability

- Provides the ability to create, clone, and deploy new VMs rapidly
- Easily add or remove VMs without the need for additional physical hardware

More efficient IT operations

- Streamlines software updates, maintenance, and network management
- Improves backup and disaster recovery through fast failover to standby VMs

Virtualization - Advantages

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Running on physical hardware

Customer-facing
Application



*These physical servers
need to be upgraded
to the next version*

Downtime = Lost \$\$\$

Running on virtualized hardware

Customer-facing
Application

*Cutover to new
servers*



Tear down old servers



*Spin up new virtual
servers and upgrade*

No Downtime

Virtualization - Disadvantages

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Upfront costs and Complexity

- Initial setup can require investment in virtualization infrastructure, including robust hardware, storage, and licensing for hypervisors or management tools
- Requires planning and technical expertise to configure and optimize virtual environments effectively

Performance Overhead

- Certain high-performance or latency-sensitive workloads may still benefit from dedicated physical hardware
- Resource contention between virtual machines can affect performance if not properly managed

Licensing and Compliance Challenges

- Licensing models can be complex in virtual environments, especially when vendors tie costs to physical cores
- Tracking and managing compliance across dynamic virtualized systems can be more difficult

Security and Isolation Risks

- Virtual environments introduce new attack surfaces, including hypervisor vulnerabilities and VM sprawl
- Ensuring strong isolation between VMs and consistent patching across hosts is critical to maintaining security

Summary

- ▶ Virtualization requires thoughtful setup and planning, but delivers significant long-term benefits
- ▶ Enables flexible, scalable configurations to support a wide range of workloads
- ▶ Provides a powerful foundation for resource sharing, isolation, and dynamic provisioning
- ▶ In cloud environments, providers manage much of the underlying complexity, allowing customers to focus on their applications rather than the infrastructure



virtualization

virtualization

Create a Virtual Server

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- ▶ Go to myCourses **Assignments > Activity – Create a Virtual Server with EC2**
 - ▶ Make sure you read and follow the directions closely
 - ▶ Take a screen shot of your running instance and upload to myCourses

